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Short communication

A 2082 Ma radiating dyke swarm in the Eastern Dharwar Craton, southern India and its implications to Cuddapah basin formation

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ABSTRACT

Using consistent paleomagnetic data together with precise Pb–Pb baddeleyite ages, on a series of mafic dykes occurring over an area of at least 70,000 km^2, a 2081.8 ± 1.1 Ma (weighted mean of 4 dykes) dyke swarm was identified intruding the Archean basement rocks in the Eastern Dharwar Craton skirting the Cuddapah basin on its north, northwest and western flanks. The geometry of these dykes collectively, due to their progressive variation in trend from N134° W to N28° E, defines a fan angle of about 162° and forms a spectacular radiating swarm converging toward a focal point beneath the Cuddapah basin. Anisotropy of magnetic susceptibility (AMS) investigations on these dykes, based on the orientation of principal eigenvectors indicate magma flow was vertically upward in them, suggesting the magma source of these dykes to be proximal to the sampling sites. These features together with reported geophysical evidence for high density material below the Cuddapah basin suggest that this mafic volcanic province probably formed due to the impact of an asthenospheric mantle upwelling perhaps triggered by a plume head or other causal mechanisms like global warming of mantle or small-scale instability like edge-driven convection. This may have resulted in the domal uplift of the continental lithosphere, large-scale crustal extension and thinning followed by thermal relaxation and subsidence that may have been responsible for the formation of the intra-cratonic Cuddapah basin, shortly after 2082 Ma.

Dyke swarms of this age (Fort Frances dykes; 2076 + 5/−4 Ma) or of slightly younger (Lac Esprit dykes; 2069 ± 1 Ma) and older (Cauchon lake dykes; 2091.1 + 1.8/−2.1 Ma) ages are fairly wide spread in the Superior province. However, a reconstruction of the paleopositions of Dharwar (Cuddapah dykes: 38° N; 180° E, A_{95} = 4°) and Superior at ~2080 Ma using paleomagnetic data (Fort Frances dykes: λp: 43° N; Lp: 184° E) does not suggest a close proximity for these provinces at that time. Their disparate locations could therefore suggest these were two distinct nodes of wide spread magmatism between 2080 Ma and 2065 Ma.

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1. Introduction

The Dharwar Craton of South India consists of two sub-blocks. The older Western Dharwar Craton (WDC: 3.3–2.7 Ga), which mainly comprises of a tonalite–trondhjemite–granodiorite (TTG) gneissic basement overlain by greenstone belts, and the younger Eastern Dharwar Craton (EDC: 3.0–2.5 Ga) made up of Late Archean (2.6–2.5 Ga) granites intrusive into subordinate amounts of older (2.9–2.7 Ga) TTG gneisses (Chadwick et al., 2000, and references therein). Greenstones in the EDC are confined to small, elongated belts which may represent terrane boundaries (Krogstad et al., 1989; Chadwick et al., 2000). The northern margin of the craton is concealed by the Cretaceous Deccan volcanic pile. It is limited in the east by the Proterozoic Eastern Ghats Mobile belt and by the Southern Granulite Belt in the south.

As in many Archean blocks mafic dyke swarms are widespread in the entire Dharwar craton, but are more prolific in the EDC. These dykes range in age from Paleoproterozoic (French and Heaman, 2010; Halls et al., 2007; Kumar et al., 2012a,b, 2014) to Late Cretaceous (Kumar et al., 2001) and have been described in detail earlier (e.g. Halls, 1982; Murthy et al., 1987; Halls et al., 2007; French and Heaman, 2010). Of these, the most dominant is the EW to ENE–WSW trending giant radiating dyke swarm emplaced between 2368.5 ± 2.6 and 2365.4 ± 1.0 Ma with an aerial extent of nearly the entire Eastern Dharwar Craton (U–Pb baddeleyite ages, Kumar et al., 2012a; French and Heaman, 2010). Other dyke swarms include a N–S oriented swarm at 2220.5 ± 4.9 Ma, a NW–SE striking swarm at 2209.3 ± 2.8 Ma, two radial swarms, one a WNW–ESE to NW–SE, 2180.8 ± 0.9 to 2176.5 ± 3.7 Ma (French and Heaman, 2010) swarm and a second NE to NW striking 2081 Ma swarm

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(Demirer, 2012). These dykes are overlain by the Proterozoic intracratonic sedimentary basins, the Kaladgi, Bhima and Cuddapah basins. The formation of these basins is highly conjectural. The occurrences of Paleoproterozoic dykes in the region lead several investigators (Bhattacharji, 1981; Kumar and Bhalla, 1983; Bhattacharji and Singh, 1984; Nagaraja Rao et al., 1987) to suggest a possible tectonic correlation between mafic magmatism and large scale crustal extension leading to basin formation in the region. We present detailed paleomagnetic, precise Pb–Pb baddeleyite age and anisotropy of magnetic susceptibility (AMS) determinations on a set of 2082 Ma radiating dykes intruding the basement rocks on the northern, north-western and western periphery of the Cuddapah basin (extending below the oldest sedimentary successions) with their focus under it.

2. Geology and sampling

The Cuddapah basin (Fig. 1) situated in the EDC is one of the largest (spreads over an area of about 44,500 km^2) Proterozoic, intra-cratonic sedimentary basins in India. During the Meso-Neoproterozoic Eastern Ghat Orogeny it was deformed into a crescent shaped basin (Goodwin, 1996). It is infilled by more than

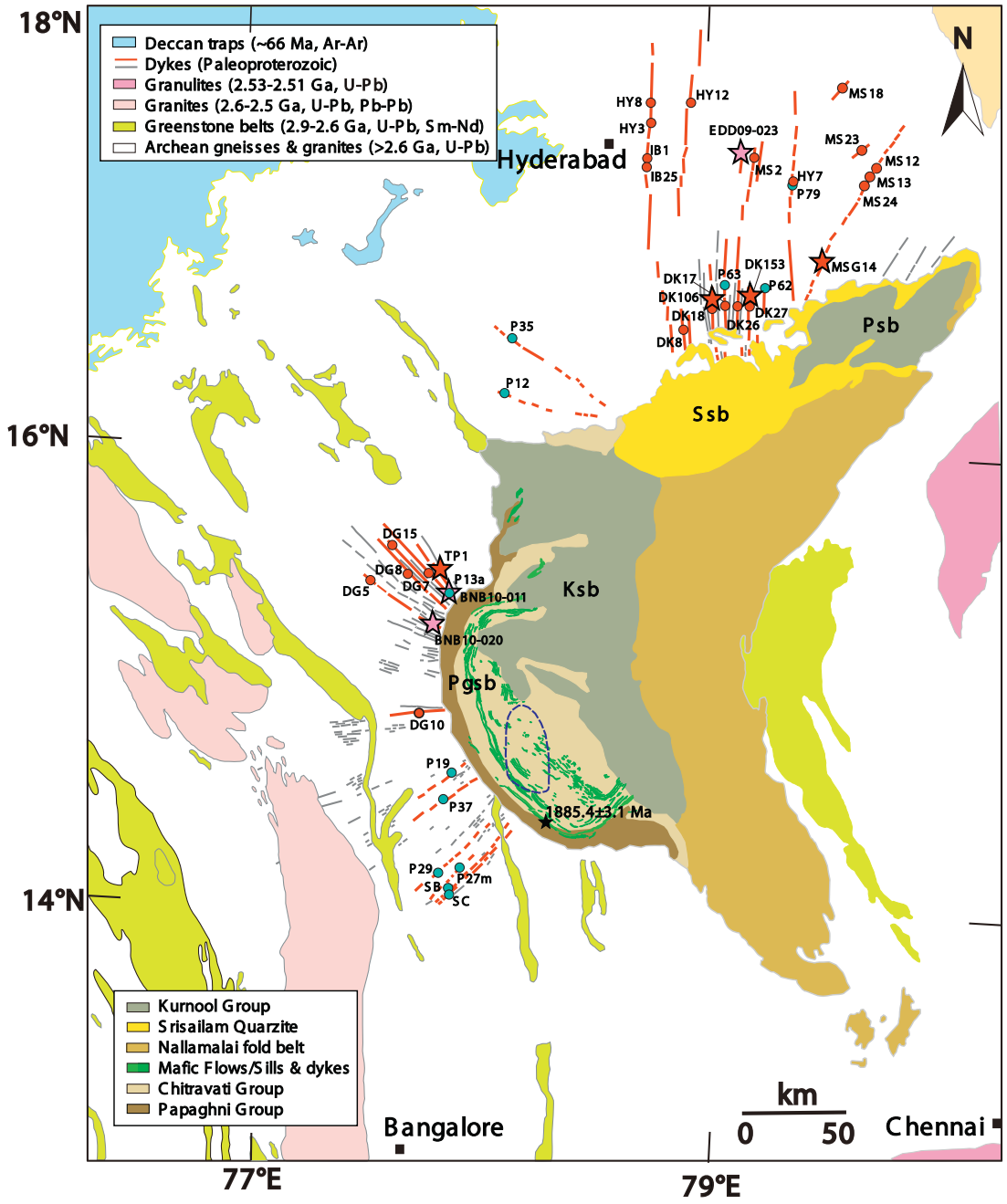


Fig. 1. Simplified geological map of the Dharwar craton showing the distribution of 2082 Ma dykes and locations of sampled sites. Red dots represent sites of paleomagnetic studies in the present study and blue dots are sites published earlier (references in Table 2). Site numbers are keyed to Table 2. Pb–Pb baddeleyite geochronological studies were performed on sites represented by red stars, pink stars are U–Pb ages reported by Demirer (2012). The black star shows the site location of the dated Pulivendula sill (French et al., 2008). Dykes shown in red belong to the 2082 Ma dyke swarm as inferred from the age and paleomagnetic data (presented here). Dykes in gray are tentatively inferred to belong to the 2082 Ma swarm, based on their field criteria (including strike pattern and cross cutting relationship with adjacent dykes). Dashed blue line demarcates the −30 mGal Geoidal corrected Bouguer contour, after Singh and Mishra (2002). Pgsb: Papagni sub-basin; Ksb: Kurnool sub-basin; Ssb: Srisailam sub-basin; Psb: Palnad sub-basin. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

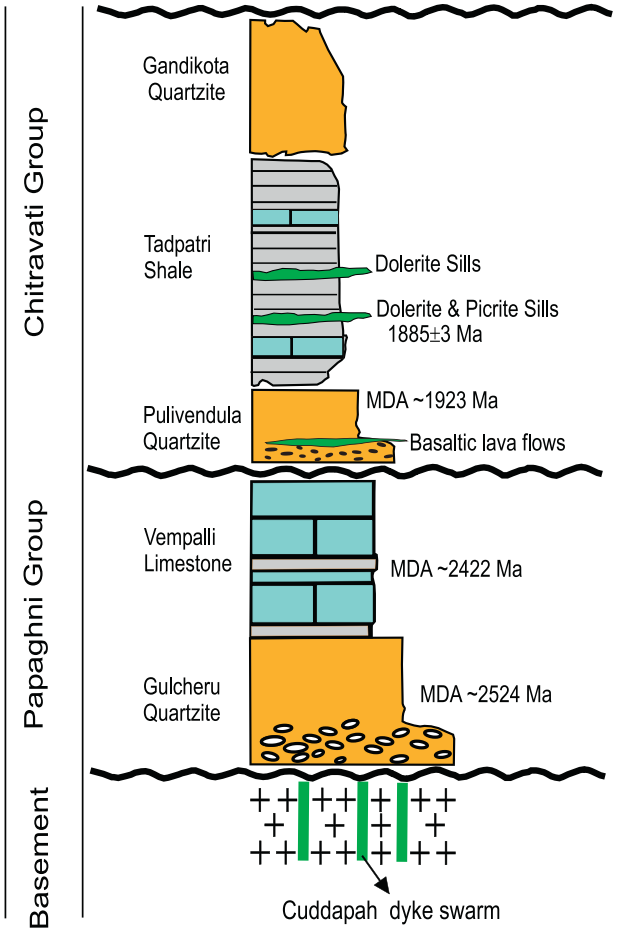


Fig. 2. Lithostratigraphy of the Papaghni and Chitravathi groups, Cuddapah basin. After Saha and Patranabis-Deb, 2014. MDA: maximum depositional age after Collins et al. (2014), age of sills after French et al. (2008).

10 km thick sedimentary successions which are divided into four sub-basins (Fig. 1), the Papaghni, Kurnool, Srisailam and Palnad (Nagaraja Rao et al., 1987). On the eastern part of the basin is the intensely deformed Nallamalai fold belt. The Papaghni sub-basin preserves the oldest of the Cuddapah sediments that include the Papaghni and the Chitravati groups. Lithostratigraphic subdivisions of sediments in the Papaghni sub-basin is given in Fig. 2.

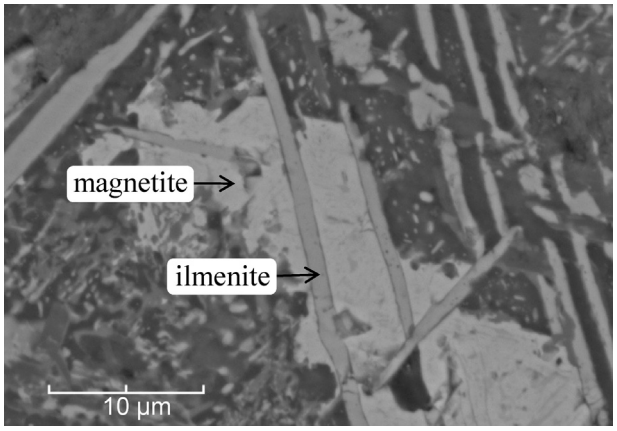


Fig. 3. Scanning electron microscope back-scattered electron image showing elongated ilmenite and Ti-poor magnetite (light gray subhedral grains seen in the background).

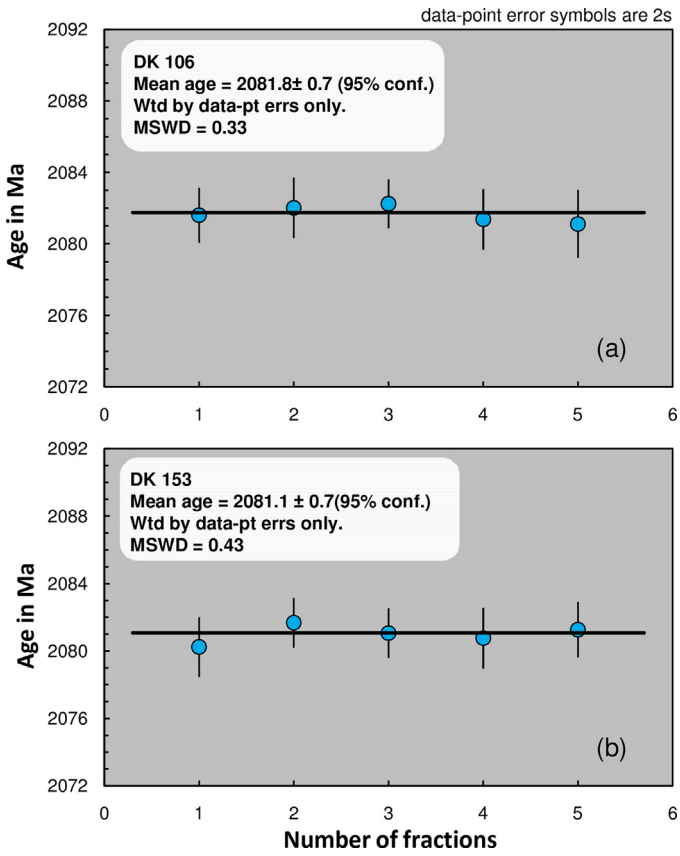


Fig. 4. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on five baddeleyite fractions each from samples (a) DK 106 and (b) DK 153 (sample location given in Fig. 1). Error bars represent 95% confidence limits of measurements.

The mafic dykes sampled for this study is from two regions (Fig. 1), one to the north of the Cuddapah basin and the second to the west of it. Both the swarms appear to intrude the Archean basement and are overlain by the Cuddapah sedimentary rocks. The northern swarm appears to be restricted to a nearly north–south trending corridor extending for at least 100 km in length, and 75 km in width. Dykes in this region have varying strike directions ranging from N75° W to N28° E (Fig. 1). Individual dyke thickness varies along strike, but is generally between 30 m and 75 m. Several of these appear to extend below the sedimentary rocks (without intruding them) in the Srisailam and Palnad sub-basins. To the west of the Cuddapah basin, dykes of this swarm are exposed for more than 100 km, skirting the basin. Like in the northern region the strike pattern of dykes in this region also varies appreciably from N134° W to N37° W (Fig. 1). Dyke thicknesses are variable from about 30–175 m. All dykes in both the northern and western sectors dip vertically, are medium to coarse-grained in their central parts and fine-grained toward the margins with sharp contacts with the country rock. Dykes in this region are overlain by sedimentary rocks of the Papaghni sub-basin. A total of 22 sites (locations given in Fig. 1) on 17 dykes were sampled from both the northern and western sectors for paleomagnetic studies. Sampling on in situ outcrops was possible from dyke margins (within 20 cm) only at 11 sites. Therefore, though paleomagnetic studies were done on samples from all the 22 sites, AMS measurements were restricted only to 11 sites. For geochronology (for baddeleyite extraction) coarse grained and differentiated portions of 4 dykes were chosen.

Experimental procedures followed for geochronological, paleomagnetic and anisotropy of magnetic susceptibility studies are described in Appendix-1.

3. Results

3.1. Petrography

Petrographic studies have been carried out on at least one sample from each site. Plagioclase and augite are the major mineral constituents, their abundances varying between 55% and 60% and 40% and 45% respectively, with minor amounts of (3–5%) of opaque minerals. Ophitic texture is very common, though in a few samples porphyritic texture was also observed. All samples are generally fresh, barring minor alteration of plagioclase in a few instances.

Scanning electron microscopy studies indicate two types of opaque grains. Interstitial medium-grained subhedral Ti-poor magnetite and elongated ilmenite (Fig. 3).

3.2. Baddeleyite geochronology

Results of Pb–Pb baddeleyite (ZrO_2) TE-TIMS (thermal extraction-thermal ionization mass spectrometry) analysis on two N–S and one NE striking dykes from the northern sector and one NW striking dyke from the western sector are given in Table 1 and Figs. 4 and 5. Sample locations are given in Fig. 1. TE-TIMS analysis of five baddeleyite fractions each from the three dykes near Neredugommu (Lat. 16.619° N, Long. 78.973° E), Puttamgandi (Lat. 16.615° N, Long. 79.114° E) and Mukundapuram (Lat. 16.831° N, Lat. 79.438° E) towns, in the northern sector yielded weighted mean Pb–Pb ages of 2081.8 ± 0.7 Ma (sample DK106), 2081.1 ± 0.7 Ma (DK153) and 2082.8 ± 0.9 Ma (MSG14) respectively and the one near Malyala (15.423° N, 77.794° E) town in the western sector gave a weighted mean age of 2081.8 ± 1.1 Ma (TP 1). All the four age determinations overlap within errors suggesting simultaneous emplacement of dykes in the two sectors, within a brief time span of not more than 4 Ma.

U–Pb ID-TIMS (isotope dilution-thermal ionization mass spectrometry) baddeleyite ages for a NNE-trending dyke in the northern sector and two NW trending dykes from the western sector were recently reported by (Demirer, 2012). Weighted mean of these three determinations is 2081.0 ± 1.6 Ma. Identical within error to the four Pb–Pb ages mean of 2081.8 ± 1.1 Ma. A weighted mean of the four age determinations of this study, together with the three determinations by Demirer (2012) gives an age of 2081.6 ± 0.4 Ma, which is here considered as the best age estimate for the emplacement of this dyke swarm.

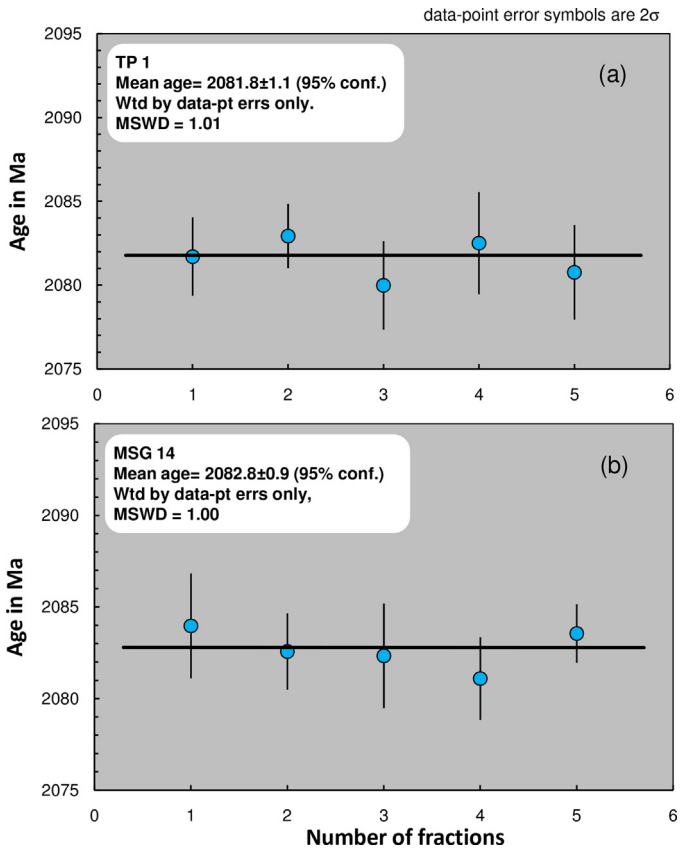


Fig. 5. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on five baddeleyite fractions each from samples (a) TP 1 and (b) MSG 14 (sample location given in Fig. 1). Error bars represent 95% confidence limits of measurements.

3.3. Paleomagnetism

A total of 221 samples, 180 from the northern and 41 from the western regions respectively from 17 dykes (22 sites) were used for paleomagnetic investigation. Results are given in Table 2 and plotted in Fig. 6. At least five samples from each site were subjected to detailed stepwise AF and/or thermal demagnetization, to a maximum of 150 mT or 600 °C, in order to identify and quantify magnetic components (Fig. 7). A high coercivity (or high

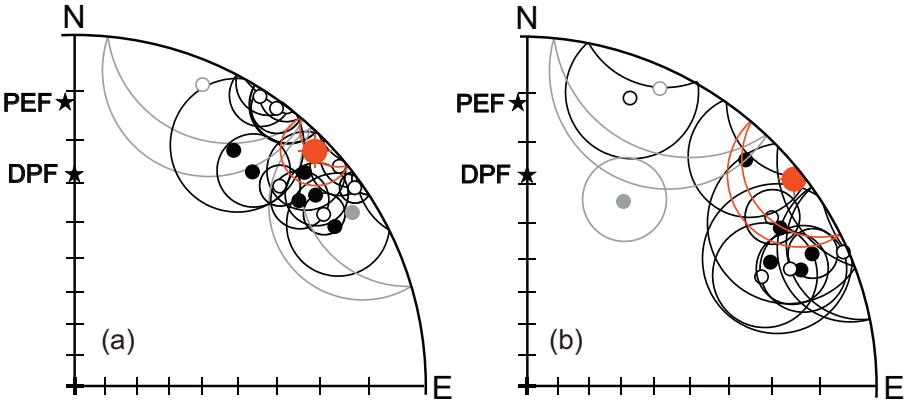


Fig. 6. Stereoplots showing paleomagnetic data. (a) Site mean characteristic remanence directions with ovals of 95% confidence of 13 dykes from the northern sector and (b) 10 dykes from the western sector. Present study and accepted published site means are represented as black circles. Gray circles are outlier data not considered for calculating the mean direction (Table 2). Red closed circle represents the grand mean of all accepted data for the respective sectors. Black stars represent DPF, Dipole field and PEF, Present Earth's field direction based on the 1995 IGRF. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
TE-TIMS Pb isotopic data on baddeleyite fractions from the Cuddapah dyke swarm samples.

Sample no.	Sample wt.	No. bloc.	$^{206}Pb/^{204}Pb$ (m)	Abs. err	$^{207}Pb/^{206}Pb$ (m)	$^{207}Pb/^{206}Pb$ (c)	Age (Ma)
DK 106							
DK 106-1	2.8	26	2906	46	0.133358 ± 106	0.128796 ± 30	2081.6 ± 1.5
DK 106-2	2.8	21	22,300	412	0.129435 ± 039	0.128827 ± 41	2082.0 ± 1.7
DK 106-3	3.3	19	32,045	211	0.129260 ± 017	0.128843 ± 17	2082.3 ± 1.4
DK 106-4	3.3	24	23,246	1094	0.129355 ± 045	0.128780 ± 41	2081.4 ± 1.7
DK 106-5	3.3	14	16,019	756	0.129616 ± 064	0.128760 ± 56	2081.1 ± 1.9
Weighted mean age = 2081.8 ± 0.7 Ma, MSWD = 0.33							
DK 153							
DK153-1	5.2	22	20,622	499	0.129169 ± 071	0.128697 ± 46	2080.2 ± 1.8
DK153-2	4.6	18	21,992	234	0.129397 ± 076	0.128802 ± 25	2081.7 ± 1.5
DK153-3	4.5	24	20,234	699	0.129315 ± 070	0.128757 ± 25	2081.1 ± 1.5
DK153-4	4.5	18	21,813	356	0.129351 ± 048	0.128735 ± 49	2080.8 ± 1.8
DK153-5	4.5	19	23,214	190	0.129408 ± 093	0.128772 ± 37	2081.3 ± 1.6
Weighted mean age = 2081.1 ± 0.7 Ma, MSWD = 0.43							
TP 1							
TP 1-1	5.3	19	43,651	1523	0.129095 ± 088	0.128804 ± 88	2081.7 ± 2.3
TP 1-2	5.3	22	26,478	392	0.129401 ± 058	0.128894 ± 57	2082.9 ± 1.9
TP 1-3	4.6	21	26,679	944	0.129162 ± 111	0.128679 ± 110	2080.0 ± 2.6
TP 1-4	5.0	19	64,597	2861	0.129079 ± 150	0.128864 ± 139	2082.5 ± 3.0
TP 1-5	5.0	20	91,564	4338	0.128911 ± 128	0.128736 ± 122	2080.8 ± 2.8
Weighted mean age = 2081.8 ± 1.1 Ma, MSWD = 1.01							
MSG 14							
MSG 14-1	4.9	17	17,200	291	0.129733 ± 132	0.128772 ± 126	2083.9 ± 2.8
MSG 14-2	4.9	18	20,344	513	0.129559 ± 071	0.128867 ± 69	2082.5 ± 2.1
MSG 14-3	4.9	17	18,439	486	0.129500 ± 129	0.128851 ± 125	2082.3 ± 2.8
MSG 14-4	5.6	19	9594	546	0.130221 ± 133	0.128760 ± 81	2081.1 ± 2.2
MSG 14-5	5.6	30	5852	376	0.131477 ± 152	0.128939 ± 34	2083.6 ± 1.6
Weighted mean age = 2082.8 ± 0.9 Ma, MSWD = 1.00							

Sample wt. is in micrograms. $^{206}Pb/^{204}Pb$ (m) and $^{207}Pb/^{206}Pb$ (m) are measured values (fraction means with standard error) and $^{207}Pb/^{206}Pb$ (c) is the corrected value. Uncertainties are $2\sigma_m$ and refer to the least significant digits. Age errors include fractionation uncertainty of 0.055%. Weighted mean ages were calculated using Ludwig (2003).

blocking temperature) component defines a well grouped characteristic magnetization direction, with northeasterly declination and very shallow inclination (Table 2). Within errors, all the sites regardless of their location in the dyke swarm have similar characteristic remanent magnetization directions (Table 2 and Fig. 6), despite appreciable variation in strike within the swarm. Sites with mean values having $\alpha_{95} > 15^\circ$ have been rejected and therefore not included in the grand mean calculations. Seventeen sites from the northern sector representing 12 dykes which vary in strike from NE to NNW yield a mean direction ($D = 47^\circ$, $I = 2^\circ$, $\alpha_{95} = 6^\circ$, $N = 12$). This direction is similar within errors to the characteristic remanence directions ($D = 52^\circ$; $I = 0^\circ$; $\alpha_{95} = 28^\circ$, $N = 4$) obtained on the four NW to SW trending dykes from the western sector. Site DG 15 was excluded from the mean calculation as it appears to be overprinted by a secondary component of an unknown younger event, as indicated by the AF demagnetization Zijderveld plot (Fig. 7f).

Several dykes with similar strike pattern and magnetization direction have been reported earlier (Belica et al., 2014; Piispa et al., 2011; Radhakrishna et al., 2013) from the periphery of the Cuddapah basin from both the northern and western sectors. These have also been included in Table 2 for easy reference. However, following our acceptance criteria ($\alpha_{95} < 15^\circ$), only three sites from the northern sector and six sites from the western sector were accepted and their locations given in Fig. 1 and data from two sites (P12, P62) from the northern sector and two sites (DG15, P13a) from the western sector were rejected (see Table 2). Total of 13 dykes from the northern sector yield a mean $D = 46^\circ$ and $I = 3^\circ$ ($\alpha_{95} = 6^\circ$) and 10 dykes from the western sector have a mean of $D = 57^\circ$ and $I = 0^\circ$ ($\alpha_{95} = 11^\circ$), and overlap within errors. All together 30 sites on 23 dykes from this swarm, including dykes from both the northern and western sectors yields a grand mean direction of $D_m = 51^\circ$, $I_m = 1^\circ$ ($\alpha_{95} = 6^\circ$) with a corresponding VGP at 38° N and 180° E ($A_{95} = 4$) (Table 2 and Fig. 8).

We were unable to obtain baked contact samples during this study to prove the primary nature of the remanence direction obtained here. However, Belica et al. (2014) have recently reported a positive baked contact test for a dyke from this (2082 Ma) swarm, in the western sector, at site P27 m (Fig. 1). Further, several other publications have also reported positive baked contact tests (Belica et al., 2014; Dash et al., 2013; Halls et al., 2007; Kumar and Bhalla, 1983) and the preservation of dual polarity (Belica et al., 2014; Radhakrishna et al., 2013) from the older 2367 Ma dyke swarm in the EDC. These observations suggest that rocks in this region have not been heated beyond their blocking temperature (~450–550 °C) after 2367 Ma. We are therefore of the opinion that the nature of magnetization recorded by the 2082 Ma dykes reported here is primary, and acquired at the time of their emplacement.

3.4. Anisotropy of magnetic susceptibility

AMS measurements were made on 121 samples from eleven sites (representing eleven dykes), nine from northern and two from western sectors. Results from all eleven sites are given in Table 3. Equal area projections of all sites are given in Figs. 9 and 10. AMS fabric in these dykes was considered primary as these dykes are unmetamorphosed and all mineral phases are fairly fresh (particularly the opaque minerals, as they record consistent remanence directions believed to be primary). The low degree of anisotropy (P_j, Jelinek, 1981), which varies between 1.019 and 1.128, in both the sectors (Table 3) is less than 1.2 in all the samples measured

Table 2
Results of paleomagnetic measurements on the Cuddapah dyke swarm.

Site	Lat. (°N)	Long. (°E)	S°	N	D_m	I_m	k	α_{95}	P lat. (°N)	P long. (°E)	A_{95}
Northern sector											
MS2	17.276	79.137	12	8	52	8	136	6	38	178	4
MS12	17.216	79.657	28	8	57	−7	91	6	31	184	4
MS13	17.173	79.633	28	11	42	4	22	10	46	185	7
MS18	17.571	79.539	50	10	55	−1	349	3	33	182	2
MS23	17.295	79.683	28	9	55	−9	173	4	31	186	3
MS24	17.144	79.618	28	10	54	−2	60	6	34	183	4
DK8	16.564	78.824	355	14	47	6	32	7	42	180	5
DK17	16.648	79.017	0	10	41	15	152	4	49	176	3
DK18	16.619	78.973	0	10	33	−1	100	5	54	194	4
DK26	16.658	79.065	355	10	37	0	68	6	50	190	4
DK27	16.615	79.114	7	10	50	11	74	6	40	176	4
IB1	17.314	78.686	3	19	52	−20	99	3	32	193	3
IB25	17.293	78.679	3	10	48	−4	19	11	39	186	8
HY3	17.441	78.702	10	9	27	−10	74	6	55	207	4
HY7	17.172	79.354	352	17	37	−1	83	4	49	191	3
HY8	17.562	78.715	4	11	49	−4	164	4	38	186	3
HY12	17.546	78.887	355	4	59	8	95	10	31	175	7
P35[c]	16.520	78.050	300	5	34	12	34	13	55	182	10
P12[c,#]	16.280	78.010	285	6	58	4	18	16	31	175	11
P62[c,#]	16.720	79.180	3	5	23	−4	20	18	61	207	13
P63[c]	16.690	79.020	356	6	22	−15	37	11	58	216	8
P79[c]	17.170	79.360	351	8	16	−6	18	14	65	220	10
MS 12 + 13 + 24	17.178	79.636	28	29	50	−1	30	5	37	184	4
HY3 + 8 + IB1 + 25	17.403	78.696	8	49	46	−12	27	4	39	192	3
HY7 + P79	17.171	79.357	352	25	36	−1	65	4	50	192	3
DK17 + P63	16.669	79.019	355	16	40	13	40	7	50	179	5
Dykes mean	17.007	79.075	–	13	46	3	48	6	43	183	4
Western sector											
DG5	15.477	77.406	309	10	55	−9	88	5	32	183	4
DG7	15.391	77.761	323	10	65	6	117	5	25	172	3
DG8	15.451	77.627	322	5	20	−8	40	12	63	211	8
DG10	14.877	77.590	85	8	67	9	67	8	23	169	7
DG15[#]	15.571	77.577	314	8	28	28	36	10	64	165	6
SB[a]	14.105	77.771	46	3	65	−17	129	11	22	183	8
SC[a]	14.092	77.770	46	3	63	14	69	15	28	167	11
P27 m[b]	14.196	77.808	51	14	66	−11	39	7	22	180	5
P29[b]	14.181	77.729	44	4	67	−1	61	12	23	175	8
P19[c]	14.610	77.800	57	5	58	9	28	15	32	171	11
P37[c]	14.500	77.770	68	7	44	6	32	11	45	179	8
P13a[c,#]	15.350	77.820	315	4	24	−4	23	20	60	202	14
Dykes mean	14.688	77.703	–	10	57	0	20	11	32	177	8
Dykes Grand mean	15.998	78.479	–	23	51	1	27	6	38	180	4

Lat. & Long.: latitude and longitude and are in degrees; S°: is strike of the dyke in degrees from north. N: number of samples studied from each site; D_m: mean declination; I_m: mean inclination; k: precision parameter; α_{95}, A_{95} = the radius (°) of the 95% circle of confidence about the mean magnetization direction; p lat. and p long.: paleocoordinates of the pole; all statistical parameters are based on sample means.

[a] Belica et al. (2014).
[b] Piispa et al. (2011).
[c] Radhakrishna et al. (2013).
[#] Sites which are not considered for mean calculations.

(except one sample with a higher value of 1.322). This can be construed as an indicator that the observed magnetic fabric is primary (Hrouda, 1982), which formed during cooling and crystallization of the magma in these dykes. Bulk susceptibility (K_m) values are generally high (average: 30.6×10^{-3}) and vary from 172 to 0.64×10^{-3}, in the northern sector and in the western sector from 41.2 to 13.5×10^{-3} (average: 26.4×10^{-3}; in SI units). This can be attributed to the presence of an interstitial ferromagnetic phase such as, Ti-poor magnetite (Knight and Walker, 1988; Hargraves et al., 1991; Rochette et al., 1991).

Data from sites MS2, MS12, MS18, MS23, DK17, DK26 and DK27, all from the northern sector show identical AMS fabric (shown in Figs. 9 and 10), with K_{max} being always subvertical and K_{min} normal to the plane of the dyke. The fabric in all these sites is inferred to indicate vertical magma flow. Samples from IB1 (also from the northern sector) show an AMS pattern wherein the K_{max} and K_{int} are in the dyke plane and K_{min} perpendicular to it, but K_{max} is close to horizontal and K_{int} is vertical. This is generally interpreted as typical horizontal flow pattern. However, this pattern could also be formed due to rolling effects on large grains when K_{max} is normal to flow direction but within the dyke plane and K_{int} is parallel to the flow (Canon-Tapia, 2004), when it could also indicate vertical flow. Samples from the site DK8 (northern sector) display abnormal magnetic fabric, with K_{max} being subhorizontal and perpendicular to the dyke plane and K_{int} being subparallel and subvertical and K_{min} being along the dyke plane. This type of magnetic fabric could form either due to the single domain effect ('inverse fabric', owing to zero susceptibility along its long axis and maximum perpendicular to it, Stephenson et al., 1986), or due to the late growth of ferromagnetic minerals in a direction perpendicular to the dyke plane (Canon-Tapia, 2004), when the magma flow direction is vertical. AMS patterns in samples from sites DG5 and DG7 both from the western sector are identical (Fig. 10). K_{max} is subvertical along the dyke plane and K_{min} is normal to the dyke trend. This type of

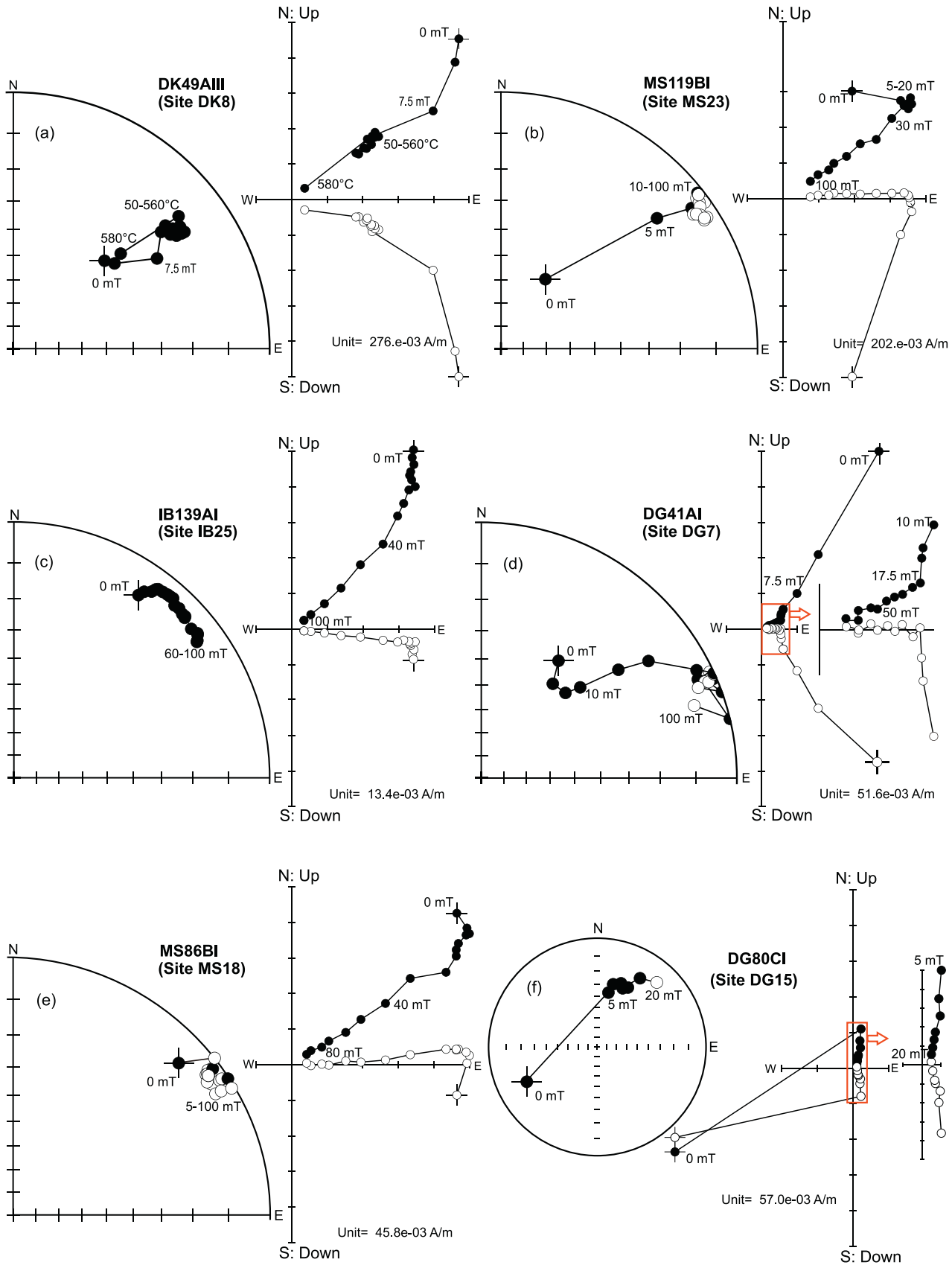


Fig. 7. Zijderveld diagrams and equal area stereonet projections showing characteristic behavior of natural remanence to thermal demagnetization (a) and AF demagnetization (b–f) for representative samples from different sites. Thermal demagnetization measurements are in °C and AF measurements are in millitesla (mT). Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0, a plotting and analysis program (Chadima and Hrouda, 2006).

Table 3
AMS directions of the Cuddapah dyke swarm.

Name	Site Lat (°N)	Site Long (°E)	Strike	Margin	K_m	L	F	P_j	T	K_{max}		K_{int}		K_{min}	
										D (°)	I (°)	D (°)	I (°)	D (°)	I (°)
Northern sector															
MS 2 site															
MS6AII	17.276	79.137	N12° E	West	22,445	1.017	1.015	1.032	−0.065	353	80	192	10	101	3
MS6AIII	17.276	79.137	N12° E	West	21,723	1.014	1.013	1.027	−0.024	325	78	198	7	107	10
MS6CI	17.276	79.137	N12° E	West	23,246	1.017	1.015	1.032	−0.065	356	83	190	6	99	2
MS6BI	17.276	79.137	N12° E	West	22,934	1.018	1.015	1.033	−0.077	335	81	196	7	105	6
MS7AI	17.276	79.137	N12° E	West	24,719	1.016	1.007	1.023	−0.369	265	69	155	7	63	20
MS7BII	17.276	79.137	N12° E	West	22,264	1.011	1.009	1.020	−0.135	249	67	145	6	53	22
MS7BIII	17.276	79.137	N12° E	West	22,997	1.015	1.007	1.023	−0.382	227	62	321	2	52	28
MS7CI	17.276	79.137	N12° E	West	24,757	1.010	1.009	1.019	−0.080	244	67	146	3	55	23
MS6BII	17.276	79.137	N12° E	East	22,842	1.016	1.017	1.033	0.023	27	79	195	11	285	2
MS6CII	17.276	79.137	N12° E	East	22,356	1.017	1.015	1.032	−0.061	13	80	192	10	282	0
MS7AII	17.276	79.137	N12° E	East	28,318	1.023	1.006	1.031	−0.605	94	77	4	0	274	13
MS7AIII	17.276	79.137	N12° E	East	24,612	1.017	1.005	1.023	−0.531	111	67	345	15	250	18
MS7BI	17.276	79.137	N12° E	East	25,778	1.021	1.004	1.027	−0.708	121	76	354	8	263	11
MS7CII	17.276	79.137	N12° E	East	26,203	1.014	1.003	1.018	−0.662	122	68	19	5	287	21
MS7CIII	17.276	79.137	N12° E	East	24,644	1.013	1.004	1.017	−0.547	123	60	343	24	245	17
MS 12 site															
MS56AI	17.216	79.657	N28° E	West	41,804	1.022	1.015	1.037	−0.185	260	75	42	12	134	9
MS56AIII	17.216	79.657	N28° E	West	43,815	1.004	1.044	1.054	0.814	253	76	29	11	121	10
MS57AI	17.216	79.657	N28° E	West	25,073	1.027	1.034	1.063	0.115	212	69	26	21	117	2
MS57CII	17.216	79.657	N28° E	West	27,607	1.026	1.027	1.054	0.029	256	79	22	7	113	9
MS58AI	17.216	79.657	N28° E	West	41,305	1.023	1.005	1.030	−0.630	239	69	36	20	129	8
MS56BI	17.216	79.657	N28° E	East	47,304	1.012	1.024	1.037	0.310	113	82	218	2	308	8
MS56BII	17.216	79.657	N28° E	East	40,516	1.001	1.015	1.018	0.903	200	49	34	40	298	7
MS57AII	17.216	79.657	N28° E	East	24,086	1.025	1.026	1.051	0.029	182	73	24	16	292	6
MS57CI	17.216	79.657	N28° E	East	24,059	1.013	1.044	1.060	0.528	199	64	35	25	302	6
DK58BII	17.216	79.657	N28° E	East	40,978	1.035	1.009	1.046	−0.589	118	72	211	1	302	18
MS 18 site															
MS88AII	17.571	79.539	N50° E	East	26,955	1.021	1.031	1.054	0.183	142	80	9	7	278	7
MS89AI	17.571	79.539	N50° E	East	41,066	1.093	1.011	1.116	−0.778	129	63	38	1	308	27
MS89AII	17.571	79.539	N50° E	East	30,420	1.096	1.022	1.128	−0.618	133	65	33	5	301	25
MS90AIII	17.571	79.539	N50° E	East	25,340	1.051	1.009	1.066	−0.701	155	72	4	16	272	8
MS90CII	17.571	79.539	N50° E	East	10,123	1.034	1.021	1.056	−0.232	219	85	36	6	126	0
MS91AIII	17.571	79.539	N50° E	East	79,954	1.039	1.017	1.058	−0.398	164	79	1	11	270	3
MS 23 site															
MS116AI	17.295	79.683	N28° E	East	62,051	1.027	1.031	1.059	0.058	104	70	254	18	347	10
MS116AII	17.295	79.683	N28° E	East	61,981	1.027	1.031	1.059	0.054	76	70	225	17	318	10
MS116BI	17.295	79.683	N28° E	East	44,574	1.017	1.033	1.051	0.321	96	73	253	15	345	6
MS116BII	17.295	79.683	N28° E	East	44,525	1.017	1.033	1.050	0.323	68	72	227	17	319	6
MS117BI	17.295	79.683	N28° E	East	57,316	1.018	1.027	1.046	0.200	93	68	261	22	353	4
MS117BII	17.295	79.683	N28° E	East	57,279	1.018	1.027	1.046	0.191	65	68	233	21	325	4
MS118BII	17.295	79.683	N28° E	East	53,981	1.003	1.027	1.034	0.782	80	64	253	26	345	3
MS118CII	17.295	79.683	N28° E	East	57,296	1.007	1.024	1.032	0.567	68	60	233	30	327	7
MS116AI	17.295	79.683	N28° E	West	57,320	1.115	1.183	1.322	0.216	303	88	185	1	95	2
MS116BII	17.295	79.683	N28° E	West	62,782	1.027	1.028	1.056	0.018	245	73	41	16	133	7
MS118AI	17.295	79.683	N28° E	West	43,762	1.010	1.033	1.045	0.548	1	89	240	0	150	1
MS118AII	17.295	79.683	N28° E	West	46,530	1.013	1.032	1.047	0.439	92	86	268	4	358	0
MS118AIII	17.295	79.683	N28° E	West	44,309	1.014	1.036	1.053	0.444	63	84	241	6	331	0
MS118AIV	17.295	79.683	N28° E	West	46,500	1.013	1.032	1.047	0.421	68	87	239	3	329	1
MS118BI	17.295	79.683	N28° E	West	56,910	1.006	1.024	1.033	0.583	302	83	93	6	183	4
MS118BIII	17.295	79.683	N28° E	West	56,877	1.006	1.024	1.032	0.620	271	82	66	8	156	4
MS119AII	17.295	79.683	N28° E	West	61,927	1.022	1.030	1.052	0.148	333	88	214	1	124	2
MS119BI	17.295	79.683	N28° E	West	52,880	1.021	1.033	1.055	0.217	45	83	210	7	300	2
MS119BII	17.295	79.683	N28° E	West	55,345	1.020	1.033	1.054	0.231	32	76	211	14	301	0
DK 8 site															
DK49AII	16.564	78.824	N355° E	East	142,365	1.011	1.009	1.021	−0.094	235	27	129	28	1	50
DK49BII	16.564	78.824	N355° E	East	137,757	1.010	1.014	1.024	0.166	219	25	113	32	340	48
DK52AI	16.564	78.824	N355° E	East	110,135	1.014	1.007	1.021	−0.368	256	24	89	66	348	5
DK52AII	16.564	78.824	N355° E	East	171,541	1.019	1.006	1.027	−0.540	248	17	94	71	340	8
DK53BII	16.564	78.824	N355° E	East	131,108	1.010	1.011	1.021	0.055	226	17	127	27	344	58
DK54AII	16.564	78.824	N355° E	East	133,261	1.010	1.010	1.020	−0.033	239	27	111	50	344	27
DK54BII	16.564	78.824	N355° E	East	103,645	1.002	1.018	1.022	0.767	261	23	119	62	358	16
DK 17 site															
DK95AI	16.648	79.017	N–S	East	775	1.002	1.002	1.003	−0.048	66	80	205	8	295	7
DK97AI	16.648	79.017	N–S	East	823	1.003	1.001	1.005	−0.557	23	69	198	21	289	2
DK97AII	16.648	79.017	N–S	East	1036	1.006	1.002	1.009	−0.589	56	61	208	26	304	12
DK97BI	16.648	79.017	N–S	East	644	1.003	1.003	1.006	−0.028	8	66	194	24	103	2
DK98BI	16.648	79.017	N–S	East	1034	1.006	1.000	1.008	−0.861	53	57	221	33	315	5
DK98BII	16.648	79.017	N–S	East	1010	1.007	1.002	1.009	−0.510	48	48	200	38	302	14
DK98BIII	16.648	79.017	N–S	East	1008	1.007	1.001	1.009	−0.639	44	51	198	37	298	13

Table 3 (*Continued*)

Name	Site Lat (°N)	Site Long (°E)	Strike	Margin	K_m	L	F	P_j	T	K_{max}		K_{int}		K_{min}	
										D (°)	I (°)	D (°)	I (°)	D (°)	I (°)
DK100AI	16.648	79.017	N–S	East	1128	1.006	1.001	1.007	−0.837	70	73	233	17	324	5
DK 26 site															
DK140AI	16.658	79.065	N355° E	West	22,797	1.02	1.023	1.043	0.073	269	58	13	9	108	31
DK141AI	16.658	79.065	N355° E	West	21,661	1.017	1.016	1.033	−0.029	274	56	31	17	131	28
DK141AII	16.658	79.065	N355° E	West	23,533	1.015	1.015	1.030	0.001	263	45	6	12	107	42
DK141AIII	16.658	79.065	N355° E	West	21,514	1.017	1.015	1.033	−0.071	274	56	33	18	132	28
DK141BI	16.658	79.065	N355° E	West	20,677	1.019	1.019	1.038	−0.010	270	60	14	8	108	29
DK 27 site															
DK149AI	16.615	79.114	N7° E	East	11,049	1.009	1.012	1.021	0.150	149	76	44	4	313	14
DK149AII	16.615	79.114	N7° E	East	8130	1.015	1.004	1.019	−0.598	139	63	355	22	259	14
DK149AIII	16.615	79.114	N7° E	East	8106	1.014	1.003	1.018	−0.616	136	62	353	23	256	15
DK149BI	16.615	79.114	N7° E	East	9283	1.021	1.018	1.039	−0.064	153	71	353	18	261	6
DK150BI	16.615	79.114	N7° E	East	5174	1.019	1.003	1.024	−0.702	108	67	298	23	207	4
DK150BII	16.615	79.114	N7° E	East	5134	1.018	1.004	1.023	−0.641	107	66	293	24	202	3
DK152BI	16.615	79.114	N7° E	East	23,819	1.017	1.001	1.020	−0.896	164	70	352	20	262	3
DK152BII	16.615	79.114	N7° E	East	23,807	1.016	1.002	1.020	−0.821	154	69	7	18	274	11
IB 1 site															
IB6AI	17.314	78.686	N19° E	West	1092	1.01	1.011	1.021	0.027	15	1	271	86	105	4
IB6AI	17.314	78.686	N19° E	West	1090	1.011	1.01	1.021	−0.068	11	0	277	85	101	5
IB6AII	17.314	78.686	N19° E	West	1147	1.013	1.007	1.021	−0.299	190	0	281	83	100	7
IB6AIII	17.314	78.686	N19° E	West	1083	1.008	1.009	1.018	0.057	15	4	265	79	105	10
IB6AIII	17.314	78.686	N19° E	West	1150	1.013	1.009	1.022	−0.184	191	6	323	82	100	6
IB6BI	17.314	78.686	N19° E	West	1055	1.010	1.012	1.023	0.090	186	4	307	83	95	6
IB6BII	17.314	78.686	N19° E	West	976	1.011	1.009	1.02	−0.079	190	8	310	75	98	13
IB6BIII	17.314	78.686	N19° E	West	930	1.010	1.013	1.022	0.139	189	3	287	73	98	17
IB7AI	17.314	78.686	N19° E	West	1215	1.009	1.012	1.021	0.135	185	2	286	82	95	8
IB7AII	17.314	78.686	N19° E	West	1158	1.008	1.013	1.021	0.208	13	1	280	78	103	12
IB7AIII	17.314	78.686	N19° E	West	1211	1.008	1.014	1.022	0.259	187	0	278	82	97	8
IB7BI	17.314	78.686	N19° E	West	1094	1.011	1.014	1.025	0.122	15	1	272	87	105	3
IB7BII	17.314	78.686	N19° E	West	1155	1.010	1.011	1.021	0.061	15	1	256	88	105	2
IB7BIII	17.314	78.686	N19° E	West	1057	1.006	1.011	1.017	0.247	195	15	346	73	103	8
IB7BIV	17.314	78.686	N19° E	West	1151	1.010	1.011	1.021	0.034	17	2	271	84	107	6
IB7CI	17.314	78.686	N19° E	West	1170	1.009	1.015	1.024	0.218	191	2	338	88	101	1
IB7CII	17.314	78.686	N19° E	West	1091	1.010	1.011	1.021	0.035	11	1	264	87	101	3
IB7CIII	17.314	78.686	N19° E	West	1155	1.010	1.013	1.023	0.148	11	3	254	84	101	6
IB7CIV	17.314	78.686	N19° E	West	1175	1.010	1.014	1.024	0.173	192	8	353	81	102	3
IB5AI	17.314	78.686	N19° E	East	894	1.010	1.005	1.015	−0.343	18	4	130	78	288	11
IB5AII	17.314	78.686	N19° E	East	916	1.011	1.002	1.014	−0.681	200	1	105	79	291	11
IB5AIII	17.314	78.686	N19° E	East	931	1.010	1.003	1.013	−0.580	19	3	122	79	289	11
IB5BI	17.314	78.686	N19° E	East	931	1.011	1.005	1.016	−0.407	23	6	132	73	291	16
IB5BII	17.314	78.686	N19° E	East	942	1.012	1.001	1.015	−0.835	19	5	194	85	289	1
Western sector															
DG 5 site															
DG31AI	15.477	77.406	N309° E	West	41,149	1.018	1.029	1.048	0.225	288	64	106	27	196	1
DG31AII	15.477	77.406	N309° E	West	16,432	1.032	1.034	1.067	0.027	281	58	108	32	16	3
DG31AIII	15.477	77.406	N309° E	West	24,026	1.020	1.034	1.055	0.264	289	48	103	42	195	3
DG31BI	15.477	77.406	N309° E	West	41,172	1.018	1.029	1.048	0.232	287	62	105	28	196	1
DG31BII	15.477	77.406	N309° E	West	28,937	1.032	1.037	1.070	0.078	291	75	111	15	21	0
DG31BIII	15.477	77.406	N309° E	West	34,265	1.020	1.028	1.049	0.178	288	69	106	21	196	1
DG31CI	15.477	77.406	N309° E	West	23,983	1.020	1.036	1.057	0.281	288	49	105	41	196	2
DG31CII	15.477	77.406	N309° E	West	25,930	1.018	1.041	1.062	0.378	268	60	110	28	14	10
DG31CIII	15.477	77.406	N309° E	West	25,986	1.019	1.040	1.061	0.359	268	61	109	27	14	9
DG33AI	15.477	77.406	N309° E	West	13,540	1.015	1.010	1.025	−0.223	280	54	101	36	11	0
DG33AII	15.477	77.406	N309° E	West	13,573	1.015	1.010	1.026	−0.199	282	57	101	33	191	0
DG33BII	15.477	77.406	N309° E	West	27,666	1.009	1.020	1.030	0.373	283	55	104	35	13	0
DG 7 site															
DG39AII	15.391	77.761	N323° E	West	26,995	1.015	1.02	1.036	0.142	303	59	119	31	210	2
DG39BI	15.391	77.761	N323° E	West	27,036	1.015	1.021	1.036	0.169	302	58	119	32	210	1
DG39BII	15.391	77.761	N323° E	West	35,059	1.009	1.014	1.023	0.202	325	67	109	19	204	13
DG40AI	15.391	77.761	N323° E	West	27,769	1.009	1.015	1.024	0.226	322	66	121	23	214	8
DG40BII	15.391	77.761	N323° E	West	33,498	1.011	1.010	1.021	−0.037	314	62	100	23	196	14
DG40CII	15.391	77.761	N323° E	West	18,361	1.011	1.023	1.034	0.367	314	71	113	18	205	6
DG41BII	15.391	77.761	N323° E	West	16,385	1.006	1.013	1.019	0.340	318	59	94	23	193	19

K_{max}, K_{int} and K_{min} are the maximum, intermediate and minimum susceptibility intensities, respectively; K_m: bulk susceptibility (*10^{-6} SI Units); P_j: corrected degree of anisotropy; T: Jelink's shape parameter (Jelinek, 1981); D: declination; I: inclination in degrees. L: magnetic lineation, F: magnetic foliation. All margin samples were collected within 20 cm of the dyke contact.

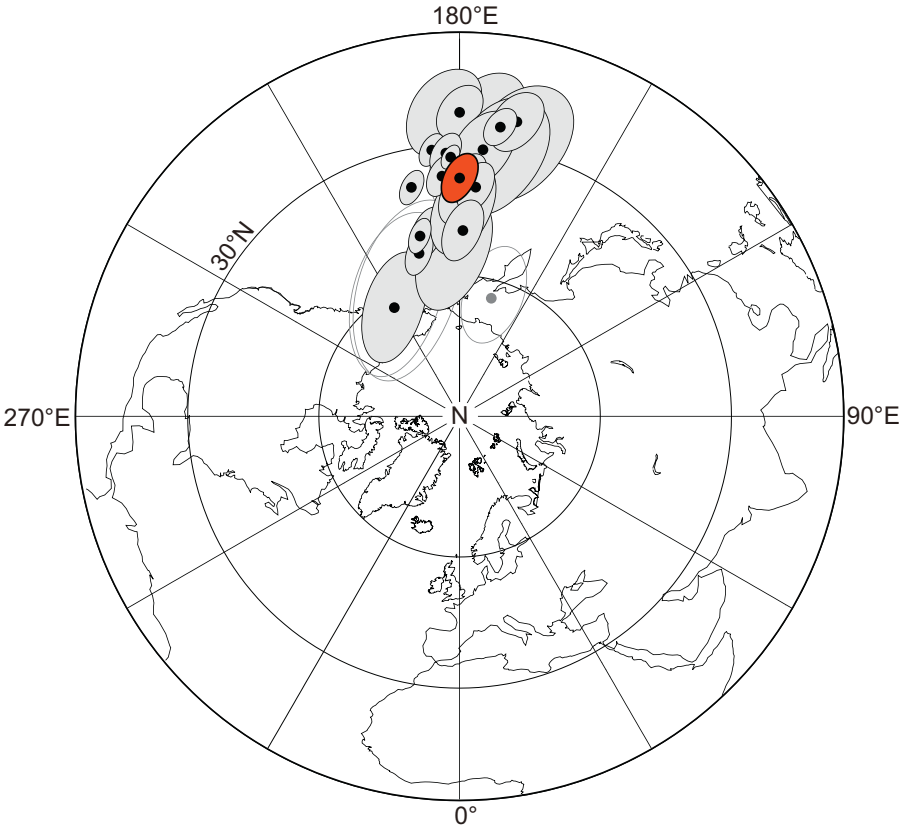


Fig. 8. Schmidt projection showing the virtual geomagnetic poles of site means from the Cuddapah dykes. Gray filled circles represent paleomagnetic poles of accepted sites. Grandmean VGP of the Cuddapah dyke swarm is shown with a filled oval of confidence in red. Shown in open gray circles are paleopoles of the outlier sites. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

fabric has also been interpreted to indicate vertical flow in dykes (Knight and Walker, 1988; Tauxe et al., 1998; Canon-Tapia, 2004).

As seen in Figs. 9 and 10, the majority of the sites show K_{max} directions from the two margins falling on either side of the dike trace. Given that, the convention is to plot AMS data in lower hemisphere projections, without exception all the western margin data plot on the western side, and the eastern margin data plot on the eastern side, this suggests that the flow was upward (Knight and Walker, 1988; Tauxe et al., 1998). We therefore infer that the AMS data obtained from chilled margins of the Cuddapah dike swarm indicate the magma flow direction in them was nearly vertical and upward.

4. Discussion

4.1. Implication to Cuddapah basin formation

Until now, several models have been proposed for the origin of the Cuddapah basin. These include its formation due to a meteorite impact (Krishna Brahmam and Dutt, 1996), by peripheral foreland subsidence (Singh and Mishra, 2002) or to passive or active rifting (Bhattacharji and Singh, 1984; Nagaraja Rao et al., 1987; Ravikant et al., 2014). Evidences put forward for the meteorite hypothesis, was the prevalence of a large oval-shaped positive gravity anomaly, caused by the existence of high density mafic material beneath the south western part of the basin and the occurrence of intense dyke swarms ascribed to impact shattering of the region. But the lack of evidence of shocked quartz in the country rocks and other features typical of meteorite impacts renders this model unlikely. Geophysical investigations in and around the Cuddapah basin have led some workers (Singh and Mishra, 2002) to infer the low and high gravity anomalies to reflect a thick sedimentary pile below the eastern Cuddapah basin and a high density ridge-like structure east of it. These evidences together with the occurrences of an ophiolite complex (Kandravolcanics, Leelanandam, 1990) in this region were interpreted as typical collision zone features and the basin formation in a peripheral foreland region. However, this is at odds with the more recently published age (~1850 Ma, U–Pb zircon) for the Kandra ophiolites (Vijaya Kumar et al., 2010). The ophiolites are coeval or slightly younger than the emplacement age of Pulivendula sills (1885 ± 3 Ma, French et al., 2008) which postdate the basal sediments (Gulcheruand Vempalli formations) within the Cuddapah basin. Therefore, indicating that the inferred collision occurred subsequent to the formation of the basin. Moreover, neither the nature of sedimentary fill in the basin nor evidence from structural investigations for convergent margin tectonic setting (listed by Ravikant et al., 2014) supports the foreland basin model. Both passive and active stretching models have also been proposed for the initiation of the Cuddapah Basin. Based on geochemical modeling of the Pulivendula–Tadpatri sill complex (1885 ± 3 Ma, French et al., 2008), Anand et al. (2003) estimated mantle potential temperatures of 1500 °C beneath the Cuddapah basin at that time, which they inferred was adequate to promote lithospheric stretching, mantle melting and passive rifting that lead to basin formation. Recently, Ravikant et al. (2014) reported a 1995 ± 11 Ma anorogenic metaluminous granite emplacement adjacent to the south-eastern margin of the Cuddapah Basin. Which they inferred had formed due to partial melting of tonalite–dioritic crust that was induced by asthenospheric upwelling and the formation of Cuddapah basin by active rifting. Occurrence of several large Paleoproterozoic dyke swarms around the Cuddapah basin to its north,

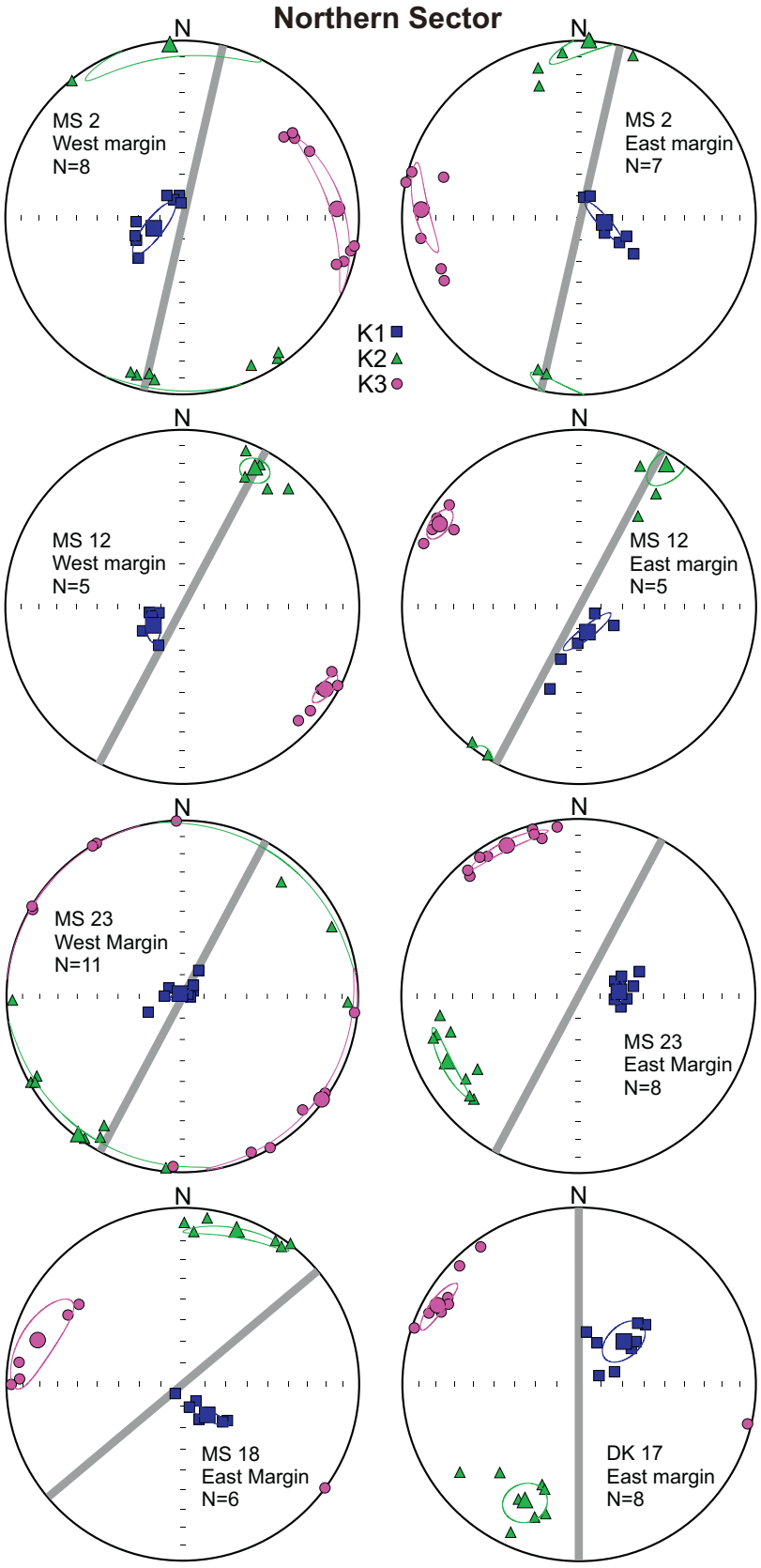


Fig. 9. Lower hemisphere projection of eigenvectors K_{max} (K_1), K_{int} (K_2), K_{min} (K_3) for representative sites showing eastern and western margin samples in separate plots from the northern sector. Dyke trends are shown as gray lines.

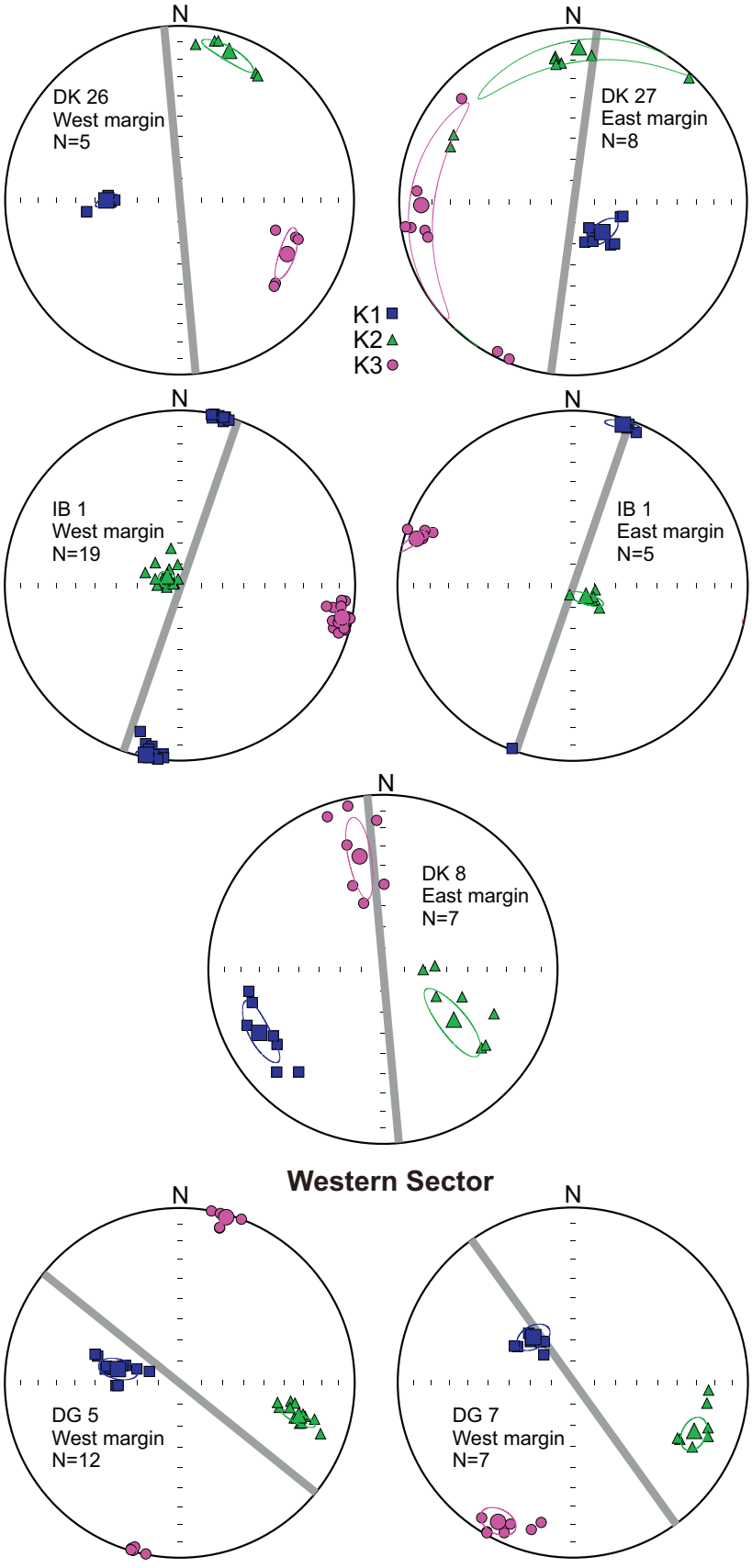


Fig. 10. Lower hemisphere projection of eigenvectors K_{max} (K_1), K_{int} (K_2), K_{min} (K_3) for representative sites showing eastern and western margin samples in separate plots from the northern and western sectors. Dyke trends are shown as gray lines.

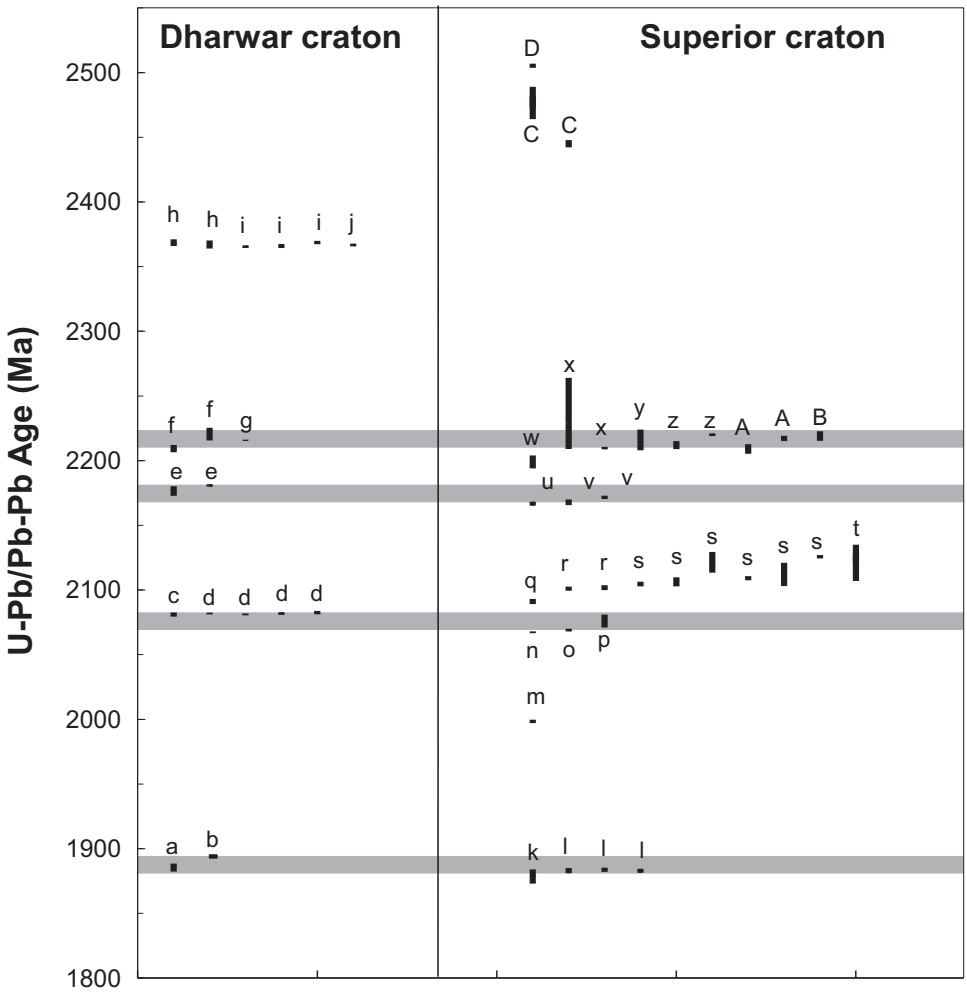


Fig. 11. U–Pb/Pb–Pb age correlation chart showing the distribution of Paleoproterozoic mafic magmatic events of Dharwar and Superior cratons. The width of individual bars corresponds to 2σ error in the respective ages. Gray band showing matching magmatic events in both cratons. Symbols correspond to the following mafic magmatic events and references: a: Pulivendula sill (1885.4 ± 3.1 Ma, French et al., 2008); b: Kamalapur dyke (~1894 Ma, Halls et al., 2007); c: Devarabanda swarm (2081.0 ± 1.6 Ma, Demirer, 2012); d: Neredugommu, Puttamgandi, Malyala and Mukundapuram dykes (2081.8 ± 0.7 Ma, 2081.1 ± 0.7 Ma, 2081.8 ± 1.1 Ma and 2082.5 ± 1.3 Ma respectively, Present study); e: Bandepalem and Dandeli dykes (2176.5 ± 3.7 Ma and 2180.8 ± 0.9 Ma, French and Heaman, 2010); f: Somala and Kandlamadugu dykes (2209.3 ± 2.8 Ma and 2220.5 ± 4.9 Ma, French and Heaman, 2010); g: AKLDyke (2215.9 ± 0.3 Ma, Kumar et al., 2014); h: Karimnagar and Hyderabad dykes (2368.5 ± 2.6 Ma and 2367.1 ± 3.1 Ma, Kumar et al., 2012a); i: Harohalli, Penukonda and Chennakottapalli dykes (2365.4 ± 1.0 Ma, 2365.9 ± 1.5 Ma and 2368.6 ± 1.3 Ma respectively, French and Heaman, 2010); j: Yeragumballi dyke (2366.7 ± 1.0 Ma, Halls et al., 2007); k: Cauchon lake dyke (1877 + 7/−4 Ma, Halls and Heaman, 2000); l: Cuthbert lake dyke, cross lake dyke and fox river sill (1883 ± 2 Ma, 1883.7 + 1.7/−1.5 Ma and 1882.9 + 1.5/−1.4 Ma respectively, Heaman et al., 1986); m: Minto dyke (1998.4 ± 1.3 Ma, Buchan et al., 1998); n: Minnesota river valley dyke (2067.3 ± 0.7 Ma, Schmitz et al., 2006); o: Lac Esprit dykes (2069 ± 1 Ma, Buchan et al., 2007); p: Fort frances dykes (2076 + 5/−4 Ma, Buchan et al., 1996); q: Cauchon lake dyke (2091.1 + 1.8/−2.1 Ma, Halls and Heaman, 2000); r: Marathon reversed dykes (2101 ± 1.6 Ma and 2101.8 ± 1.9 Ma, Hamilton et al., 2002); s: Marathon dykes (2104.6 ± 1.8 Ma, 2106.3 ± 3.5 Ma, 2121.4 + 7.8/−8.2 Ma, 2109.1 ± 1.6 Ma, 2112 ± 9 Ma and 2125.7 ± 1.2 Ma, Halls et al., 2008); t: Marathon dykes (2121 + 14/−7 Ma, Buchan et al., 1996); u: Biscotasing dykes (2166.7 ± 1.4 Ma, Buchan et al., 1993); v: Biscotasing dykes (2167.8 ± 2.2 Ma and 2171.6 ± 1.2 Ma, Halls and Davis, 2004); w: Couture dykes (2199 ± 5 Ma, Maurice et al., 2009); x: Magurie and Klotz dykes (2229 + 35/−20 Ma and 2209.7 ± 0.8 Ma, Buchan et al., 1998); y: Senneterre dykes (2216 + 8/−4 Ma, Buchan et al., 1996); z: Kogalukbay and Anuc dykes (2212 ± 3 Ma and 2220 ± 1 Ma, Maurice et al., 2009); A: Nippsing intrusions (2209 ± 3.6 Ma, and 2217.2 ± 4 Ma, Noble and Lightfoot, 1992); B: Nippsing sills (2219 + 3.6/−3.5 Ma, Corfu and Andrews, 1986); C: Matachewan dykes (2473 + 16/−9 Ma and 2445.8 + 2.9/−2.6 Ma, Heaman, 1997); D: Ptarmigan dykes (2505.3 + 2/−1.3 Ma, Buchan et al., 1998).

west and south encouraged Bhattacharji (1981), Kumar and Bhalla (1983), Bhattacharji and Singh (1984), and Nagaraja Rao et al. (1987) to invoke a thermal model (Haxby et al., 1976) for the formation of this basin. Where in magmatism causes the crust to up warp due to heating, followed by crustal thinning, subsidence and gravity faulting as a result of thermal relaxation.

Precise Pb–Pb and U–Pb baddeleyite ages and consistent paleomagnetic data on mafic dykes around the Cuddapah basin immediately to its north and west, provides unambiguous evidence for a large radiating dyke swarm (herein named the Cuddapah dyke swarm) encompassing an area of at least 70,000 km^2 in the Eastern Dharwar Craton. Dykes in the northern sector radiate from N75° W to N28° E and in the western sector between N134° W and N37° W with fan angles of 103° and 97° respectively, defining a total fan angle of about 162°, with convergence beneath the Cuddapah basin. A careful examination of the strike patterns of dykes from the northern and western sectors independently, suggests two distinct loci (Fig. 1) for these sectors. Perhaps similar to that inferred by Baragar et al. (1996) for the Mackenzie dyke swarm. Additional age and paleomagnetic data on other dykes in the swarm are needed to substantiate this inference.

Anisotropy of magnetic susceptibility, an excellent proxy for inferring petrofabric (preferred alignment of minerals) and magma flow direction in dykes (Tauxe et al., 1998; Canon-Tapia and Herrero-Bervera, 2009), indicates near vertical flow of magma in these dyke fissures. This type of flow pattern is believed to be typical of regions proximal to a magma source (<500 km, Ernst and Baragar, 1992).

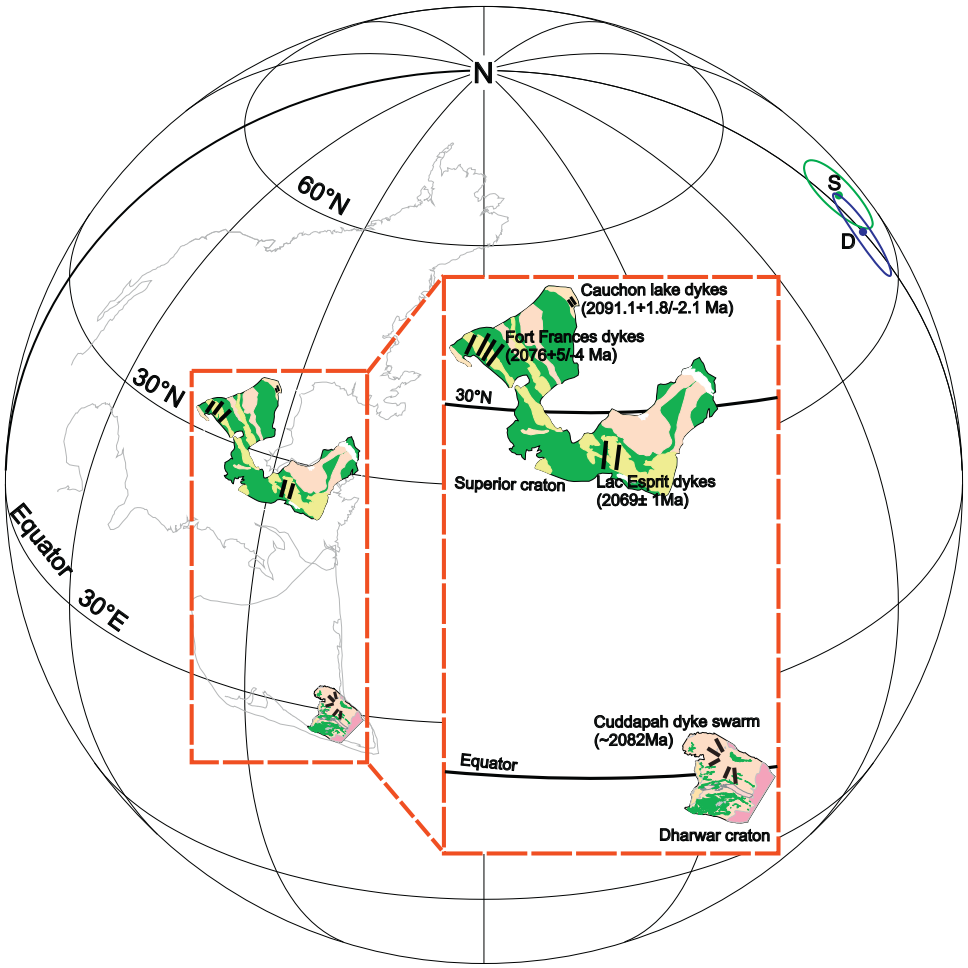


Fig. 12. Orthogonal projection showing paleopositions of the Dharwar and Superior cratons at ~2080 Ma, using VGP's of Cuddapah dyke swarm (present study) and Fort Frances dykes (Halls, 1986) respectively. Superior Craton was rotated by 80° (anticlockwise) about the Euler pole: 45 °N and 173 °E. Inset is an enlarged view of the reconstructed cratons at ~2080 Ma to display dyke swarm orientation. Dyke swarms are represented as black thick lines. Outlines of Neoarchean sequences are also shown within the cratons in green for the comparison of their regional structural grain. This reconstruction was made assuming that the magnetic field was a geocentric axial dipole during this time using "GMAP" computer program (Torsvik and Smethurst, 1999). S: Superior VGP; D: Dharwar VGP.

The large aerial extent, radiating dyke pattern and vertical magma flow of the Cuddapah swarm represents a configuration suggesting dyke propagation occurred above a centrally located magma source. Furthermore, its rapid emplacement (<4 Ma, accounting for errors in the estimates) is characteristic of starting plume head eruptions such as that represented by the Deccan basalts (<1 Ma, Baksi, 2014 and references therein) and the Siberian traps (<1 Ma, Kamo et al., 2003). In the zone of foci of the radiating Cuddapah dyke swarm several geophysical anomalies have been reported. The occurrence of a circular gravity high (~55 mGal, Singh et al., 2004), a 100 km wide highly conductive body (resistivity <100 ohm-meter, Naganjaneyulu and Harinarayana, 2004) and seismic evidence (Chandrakala et al., 2010) for a 15–20 km thick, high velocity, high density (Vp: 7.10–7.30 km/s; density 3.07–3.16 g/cm^3) mass. All inferred to be caused by underplated magma beneath the basin. Thinning of the lithosphere beneath the Cuddapah basin (<140 km) compared to the western part of the EDC (180–200 km) through the WDC (260 km) as shown by Gupta et al. (2003) may indicate a region of stretched continental lithosphere impacted by mantle upwelling and thermal erosion. These features (assumed to be coeval with the Cuddapah dyke swarm) lend support to a model in which the Cuddapah dyke swarm originated from a mantle plume. However, other features of plume head magmatism, such as coeval volcanic and plutonic rocks (LeCheminant and Heaman, 1989; Ernst, 2014) are not known from this region. Therefore, the causal factor for the associated uplift due to thermal perturbation that resulted in the basin formation can also be explained by global warming of mantle (Coltice et al., 2009) or small-scale instability like edge-driven convection (Davies and Rawlinson, 2014). We therefore propose that the radiating Cuddapah dyke swarm was probably derived by decompressional melting of an asthenospheric mantle. The mantle upwelling could have resulted in domal uplift of the continental lithosphere, causing crustal extension and thinning followed by thermal relaxation and subsidence which may have been responsible for the formation of the Cuddapah basin.

In basin formation models that invoke thermal subsidence, the onset of sedimentation is considered to be delayed by about 60–90 Ma after the initial rifting and uplift (caused by the thermal effect of the mantle plume, as observed in the case of the intracratonic West Siberian Basin, Campbell and Griffiths, 1990; Saunders et al., 2005). The cause for this delay is attributed to the decay of thermal uplift (Saunders et al., 2005). Since we advance a similar model for the initiation of the Cuddapah basin, it is likely sedimentation in this basin commenced shortly after 2020 Ma, approximately 60 Ma after the emplacement of the Cuddapah dyke swarm at 2082 Ma (present study and Demirer, 2012). In this context the Srikalahasti granite dated at 1995 ± 11 Ma and previously

inferred (Ravikant et al., 2014) to represent the thermal event responsible for the Cuddapah basin formation may represent a later event related to subsequent evolution of the basin. Therefore, it is evident that episodic heating at ~2082 Ma (Cuddapah dyke swarm, this study), ~2000 Ma (Srikalahasti granite, Ravikant et al., 2014) and ~1880 Ma (Pulivendula sill, French et al., 2008) and alternate cooling played a vital role in the development of this basin.

4.2. Coeval dyke swarms at 2082 Ma

Coeval ~2082 Ma dyke swarms are known from the Superior province. These include the Fort Frances dykes (2076 + 5/−4 Ma, Buchan et al., 1996), Cauchon lake dykes (2091 ± 2 Ma, Halls and Heaman, 2000), and Lac Esprit dykes (2069 ± 1 Ma, Buchan et al., 2007). To verify if this was a single large event with coeval basin formation and sedimentary units spreading over the two cratons that may have been neighbors during that time, we attempted a comparison (Fig. 11) of the Paleoproterozoic dyke events ('bar code' match, Bleeker and Ernst, 2006) from these cratons and also reconstructed their paleopositions at ~2080 Ma using paleomagnetic data (Fig. 12).

Eight dyke events at ~2505 Ma, ~2460 Ma, ~2213 Ma, ~2170 Ma, ~2111 Ma, ~2075 Ma, ~1998 Ma and ~1882 Ma have been reported from the Superior during the Paleoproterozoic (Fig. 11). Mafic magmatism know from Dharwar include events at ~2367 Ma, ~2215 Ma, ~2180 Ma, ~2082 Ma and 1885 Ma. Although there appears to be a good number of matching dyke events between the two cratons at ~2213 Ma, ~2170 Ma, ~2082 Ma and ~1885 Ma, suggesting they could probably be neighbors during that time. Several events present in the Superior at ~2505 Ma, ~2460 Ma, ~2111 Ma and ~1998 Ma are unknown from the Dharwar, and also notable is the absence of the ~2367 Ma dykes in the Superior.

Using the paleomagnetic pole (λp: 38° N; Lp: 180° E) obtained for the Cuddapah dyke swarm in this study and the reported pole (λp: 43° N; Lp: 184° E) for the Fort Frances dyke swarm (Halls, 1986, since they have overlapping ages), a reconstruction of their paleopositions at about 2080 Ma shows the two cratons Dharwar and Superior were located disparately (Fig. 12) near the paleoequator and 30° N latitudes respectively. Paleomagnetic data from Cauchon lake and Lac Esprit dyke swarms (Halls and Heaman, 2000; Buchan et al., 2007) being similar to the Fort Frances dyke data, are consistent with this inference. According to this reconstruction the two provinces were separated by a minimum of about 3000 km at that time. Suggesting, a 'bar code' comparison alone without paleomagnetic data could sometimes be ambiguous. Similar inference was put forward by Kumar et al. (2012b) based on the paleomagnetic reconstruction of these two provinces at ~2215 Ma also. Based on these evidences we suggest that the ~2080 Ma magmatic events in the Superior and Dharwar probably represent independent events generated from multiple sources. Such disconnected magmatic events have been identified at 65–62 Ma, 90 Ma, 120 Ma, 133 Ma, 1115–1070 Ma, 1270 Ma, 1380 Ma, and 1460 Ma (Ernst, 2014).

5. Conclusions

A consistent paleomagnetic and precise Pb–Pb baddeleyite age determinations, led us to identify a large 70,000 km^2, 2081.6 ± 0.4 Ma old radiating dyke swarm skirting the Cuddapah basin on its north, northwest and western flanks, with its oldest sediments overlying them. The foci of this dyke swarm being beneath the Cuddapah basin and coinciding with geophysical anomalies that indicate underplated magma together with anisotropy of magnetic susceptibility data suggesting vertically upward magma flow, suggests asthenospheric mantle upwelling and mantle plume activity in this region. In a preferred model, the consequence of plume head impact was responsible for large-scale crustal extension followed by thermal relaxation and thinning, resulting in subsidence that may have been responsible for the formation of the intra-cratonic Cuddapah basin, shortly after 2082 Ma.

Though coeval ~2080 Ma dyke swarms also occur in the Superior province a paleoreconstruction of the Dharwar and Superior provinces shows these large igneous provinces were separated by more than 3000 km, suggesting that these LIPs were disparate magmatic events.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2015.05.039

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